Proof by Deduction

Exam code: 9MA0

Download PDF

Written by: Paul
Updated on 18 June 2025

Did this video help you? Yes No

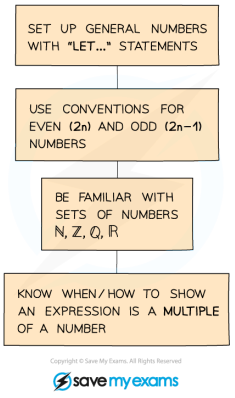
Help me understand

Proof by deduction

What is proof by deduction?

Proof by deduction is when a mathematical and logical argument is used to show whether or not a result is true

How do I use proof by deduction?

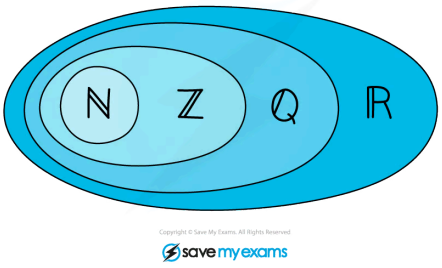


You may also need to:

- Write multiples of **n** in the form **kn** for some integer **k**
- Use algebraic techniques, showing logical steps of simplifying
- Use correct mathematical notation

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- $\mathbb{Z}$ - the set of integers
- $\mathbb{Q}$ - the set of quotients/rational numbers
- $\mathbb{R}$ - the set of real numbers



Examiner Tips and Tricks

- Try the result you are proving with a few values
  - Use a sequence of them (eg 1, 2, 3)
  - Try different types of numbers (positive, negative, zero)
- This may help you see a pattern and spot what is going on

Worked Example

Prove that the square of an even number is also even.

WRITE EVEN NUMBERS IN THE FORM $2n$

LET AN EVEN NUMBER BE $2n$

$(2n)^2 \equiv 4n^2$

$\equiv 2(2n^2)$

$\equiv 2k$ FOR SOME INTEGER k

"EVEN" IS THE SAME AS "MULTIPLE OF 2"

SO THE SQUARE OF AN EVEN NUMBER IS ALSO EVEN

Copyright © Save My Exams. All Rights Reserved
save my exams

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.